VoxClinBench: A Cross-Lingual, Cross-Disease Clinical Voice Biomarker Benchmark

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Abstract

Voice-based general-practitioner (GP) diagnosis — a clinician or AI that listens once and infers a broad range of conditions — is an aspirational clinical AI target. Two first-principles arguments ground any benchmark aimed at it. (i) Physical (laryngeal/motor) and psychiatric voice conditions foreground different acoustic substrates (articulatory/phonatory kinematics vs prosodic/spectral affect) and must be evaluated in separate families, not as a unified joint prediction. (ii) Cross-lingual evaluation is a load-bearing dimension of any voice biomarker claim: a model that performs well across languages is stronger than one that only works in its training language. Every cross-corpus row in VoxClinBench tests this directly. No existing benchmark operationalises both commitments across multiple disease families and multiple languages under a common protocol. Single-disease challenges (ADReSSo [Pan et al., 2021], AVEC [Ringeval et al., 2019], FEMH [Koudounas et al., 2024], CogBench [Feng et al., 2025], SLP-Helm [Patel et al., 2025]) cover narrow niches; prior multi-disease baselines on B2AI (MARVEL [VoxClinBench Authors, 2026]) report single-seed, single-model evaluations that hide model–disease disagreement and can produce phantom leaderboard claims on small-N psychiatric tasks. We introduce **VoxClinBench**: 18 subject-disjoint tasks (16 within-corpus + 2 cross-lingual LODO) across 12 conditions and **4 languages (English, Spanish, German, Mandarin)** over **five** public or credentialed corpora (B2AI v3.0, NeuroVoz, SVD, E-DAIC, MODMA; E-DAIC is the AVEC 2019 superset of the original DAIC-WOZ [Gratch et al., 2014] cohort plus 86 new interviews; MODMA is retained as a *cross-lingual validation target only* (n=52 Chinese, small-sample caveat; not used as a training source for depression LODO in v0.3)), with a 5-seed bootstrap + DeLong [DeLong et al., 1988, Sun and Xu, 2014] + Holm–Bonferroni protocol stack and a panel of reference models (frozen SSL across multiple backbones and layers, LP-FT, handcrafted AVQI, and an 8-modality CNN–Transformer). The benchmark reports, rather than ranks: it lists operating points for every reference model in parallel so that readers can see where languages fail to transfer, where different methods win different conditions, and where small-sample protocols mask effect sizes. Under the 5-seed patient-level bootstrap, our 8-modality reference baseline posts **macro-physical-8** $= 0.834 \pm 0.008$ (n=8 phonation-organ diseases) and **macro-psychiatric-4** $= 0.684 \pm 0.061$ (n=4 psychiatric conditions) on B2AI Tier-2, with combined macro-12 $= 0.784 \pm 0.020$ (95% CIs in supp §2); these two family-level numbers are the primary headline, reported in parallel rather than collapsed, because physical and psychiatric voice disorders are not a single skill. Representative findings that the shared protocol renders visible include a multilingual-SSL directional advantage on non-English cohorts ($+0.02$–0.04 AUROC on NeuroVoz and SVD); a disease-dependent cross-lingual split on physical tasks — B2AI→NeuroVoz Parkinson's transfers zero-shot at 0.850 AUROC while B2AI↔SVD VF-paralysis collapses to ≈ 0.46 in both

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directions; a paired GL-vs-raw causal probe on E-DAIC (N=275, Δ=+0.018) and SVD (N=513, Δ=+0.032, p<0.001) showing Griffin–Lim reconstruction does not degrade mean-pooled SSL probe AUROC on these two corpora; and a full-FT catastrophic-forgetting collapse (0.45–0.56) on small-N Griffin–Lim-reconstructed B2AI audio surfaced only by the matched frozen-probe comparator. Harness, manifests, leaderboard: `github.com/voice-bench-submission/voxclinbench`.

1 Introduction

Voice is a contact-free, low-cost biosignal carrying measurable correlates of neurological, respiratory, laryngeal, and psychiatric disease. Acoustic biomarkers have been reported for Parkinson's disease, vocal-fold pathology, airway stenosis, cognitive decline, depression, PTSD, and ADHD [Mendes-Laureano et al., 2024, Gratch et al., 2014, Marmar et al., 2019, Pan et al., 2021], yet the field is in a *reproducibility crisis of its own making*: published results are rarely comparable because every study picks its own subject-level split, its own positive/negative threshold, its own primary metric, and its own held-out cohort. Two papers reporting "AUROC 0.85 for depression from voice" on nominally the same corpus may in fact share fewer than a third of test subjects, use incompatible label definitions (PHQ-8$\geq$10 vs clinician MDD), and score on non-overlapping audio conditions. SOTA claims thus become uncheckable — a failure that is structural, not malicious. Cross-lingual generalisation — the property that should matter most for a biomarker that aspires to deployment — is almost never tested, and when it is, every author picks a different held-out cohort.

We call the implicit goal of this literature the *cross-lingual general-practitioner* (CLGP) setting — a voice model that, like a general clinician, can pick up on multiple disease families in whatever language the patient happens to speak. No single paper has claimed to deliver a CLGP, and we do not either. But progress toward it cannot be measured without a shared benchmark that evaluates both (i) across disease families with *separate* headline scores (physical and psychiatric conditions are not a unified skill), and (ii) across multiple languages with explicit cross-lingual transfer rows. VoxClinBench is our instrument for that measurement. It reports two family-level headline macros, never a single combined score as primary.

Parallel benchmarking traditions in speech and audio solve an analogous problem for *non-clinical* tasks. SUPERB [Yang et al., 2021] and HEAR [Turian et al., 2022] unified paralinguistic probes across a shared set of SSL backbones; Open ASR Leaderboard did the same for transcription. No comparable benchmark exists for clinical voice. Two concurrent efforts — CogBench [Feng et al., 2025] (cognitive impairment, English + Chinese) and SLP-Helm [Patel et al., 2025] (pediatric speech-language pathology) — each cover one disease family in ≤ 2 languages. Neither spans the physical / psychiatric divide, and neither includes a consortium-scale corpus such as Bridge2AI-Voice (B2AI) [Bensoussan et al., 2025, Goldberger et al., 2000].

We introduce **VoxClinBench**, to our knowledge the first open benchmark that unifies five clinical voice corpora across four languages and twelve disease conditions spanning both physical and psychiatric families, under a subject-level bootstrap protocol. VoxClinBench is designed to be *runnable*: splits are released as subject-ID manifests (no redistribution of restricted audio), a `voxbench` pip package ships the evaluation harness, bootstrap CI, DeLong comparison, and Holm–Bonferroni correction, and a fetch-on-use CLI verifies credentials for restricted corpora (B2AI, E-DAIC, SVD) before downloading. Psychiatric corpora at this release are English-only for training (B2AI, E-DAIC), with MODMA (n=52 Chinese) retained as a cross-lingual *validation target* only (zero-shot test, not a training source); scaled cross-lingual psychiatric evaluation remains an open roadmap item. Two reference baselines are included — MARVEL [VoxClinBench Authors, 2026] and a 7-branch CNN–Transformer over 8 modalities that uses a Griffin–Lim bridge to make B2AI's public spectrogram tier usable as an SSL input for reference comparison — along with raw per-subject predictions so any future submission can be compared head-to-head.

Contribution. VoxClinBench is organised around two first-principles commitments. (1) *Disease-family decomposition.* Family A (physical) and Family B (psychiatric) are scored independently with separate Holm–Bonferroni families, reflecting that articulatory/phonatory vs prosodic/spectral features dominate the two regimes; we do *not* claim a single model jointly predicts across families. (2) *Cross-lingual evaluation as a load-bearing dimension.* A voice biomarker that performs well across languages is stronger than one that only works in its training language; every cross-corpus

row in VoxClinBench tests this directly. The benchmark itself covers 18 subject-disjoint tasks (16 within-corpus + 2 cross-lingual LODO) across 4 languages and 12 conditions spanning physical (PD, VFP, iSGS, SD/LD, MCI, benign lesions, MTD, chronic cough) and psychiatric (depression, PTSD, ADHD, psychiatric history) families — a combined physical-plus-psychiatric scope no prior multi-disease benchmark matches, including concurrent CogBench [Feng et al., 2025] and SLP-Helm [Patel et al., 2025] (Table 1). Shipped as a pip-installable harness with subject-level bootstrap CIs, DeLong testing, and Holm–Bonferroni correction, alongside a panel of reference models reported in parallel columns without head-to-head "beat" verdicts.

Empirical consequences of the design: what the shared protocol reveals. Under matched subjects, splits, and seeds (5-seed bootstrap CIs, Holm–Bonferroni within family), the protocol surfaces the following directional patterns: (i) on non-English corpora, multilingual SSL (XLSR-53) shows $+0.02$ to $+0.04$ higher AUROC than English-only WavLM on SVD, while on NeuroVoz the two are at parity (§6(a)); (ii) cross-lingual physical-disease transfer is disease-dependent: a full ensemble trained on B2AI English PD reaches 0.850 ± 0.006 AUROC zero-shot on Spanish NeuroVoz PD (vs. MARVEL zero-shot 0.528), whereas VF-paralysis transfer between English B2AI and German SVD collapses to ≈ 0.46 AUROC in both directions (§6(b)); (iii) a paired GL-vs-raw WavLM-L9 probe shows Griffin–Lim reconstruction does not degrade probe AUROC on the SVD voice-pathology task (N=513, Δ=+0.032, p<0.001) and is directionally consistent on E-DAIC depression (N=275, Δ=+0.018, p≈0.08), reducing but not ruling out a standing concern that Griffin–Lim phase loss might impose a ceiling on downstream AUROC for other B2AI conditions where no paired raw audio is available (§6(c), Table 3). (iv) on B2AI Tier-2, end-to-end full fine-tuning of wav2vec2-base and HuBERT-base on Griffin–Lim-reconstructed audio collapses to macro AUROC 0.45–0.56 across five small-N conditions ($n_{\text{pos}} \in [38, 73]$), while frozen-SSL probes on the same subjects and seeds reach 0.79 macro; the shared protocol surfaces this catastrophic-forgetting failure that single-cohort studies cannot (§6(d)).

2 Related Work

SSL speech benchmarks. SUPERB [Yang et al., 2021] and HEAR [Turian et al., 2022] established shared-probe protocols across SSL backbones on non-clinical tasks (ASR, emotion, speaker-ID, keyword spotting, environmental audio). They fix an evaluation harness and a family of frozen-feature heads so that every new backbone is compared on identical ground. VoxClinBench imports this discipline — frozen features, fixed heads, subject-disjoint splits — into the clinical domain, where it has not previously been applied at scale.

Position vs general-purpose speech and medical benchmarks. Prior benchmark efforts cover three largely disjoint regions: general speech/audio representation (SUPERB [Yang et al., 2021], HEAR [Turian et al., 2022], ComParE [Schuller et al., 2013]), multilingual speech or NLP evaluation (FLEURS [Conneau et al., 2022], XTREME [Hu et al., 2020]), and non-speech medical tasks (MedMNIST v2 [Yang et al., 2023]). None jointly satisfy the three axes required for a clinical-voice benchmark: *clinical disease labels*, *multilingual corpora*, and *cross-lingual transfer evaluated as a first-class task* under a subject- and dataset-disjoint protocol (Table 4).

VoxClinBench is the only entry positive on all three of {Clinical, Multilingual, Cross-lingual transfer}, and the only clinical-voice benchmark that ships a leave-one-dataset-out protocol with pre-frozen subject-disjoint splits (scope comparison against SUPERB, HEAR, FLEURS, ComParE, MedMNIST v2, XTREME in supp Table 4). §3 describes how this scope is realised in the task catalogue and evaluation harness.

Single-disease clinical benchmarks. Parkinson's-only work has accumulated splits across NeuroVoz, GITA, GermanPD, CzechPD, and ItalianPVS [Mendes-Laureano et al., 2024, Escobar-Grisales et al., 2024, Adnan et al., 2025]; depression has AVEC and DAIC-WOZ/E-DAIC [Gratch et al., 2014, Ringeval et al., 2019, Bailey and Plumbley, 2021, Dumpala et al., 2021]; dementia has ADReSS/ADReSSo [Pan et al., 2021]; laryngeal pathology has FEMH and SVD [Institute of Phonetics, Universität des Saarlandes, 2007, Koudounas et al., 2024, Javanmardi et al., 2023]; cough and respiratory has large-scale corpora from the COVID era [Laguarta et al., 2020]. Each community evaluates in isolation.

Concurrent cross-lingual clinical benchmarks. CogBench [Feng et al., 2025] (cognitive impairment, en+zh) and SLP-Helm [Patel et al., 2025] (pediatric SLP) are concurrent but each scoped to one disease family in ≤ 2 languages. VoxClinBench is complementary: physical + psychiatric, 4 languages, 12 conditions (Table 1).

Table 1: Scope differentiation among concurrent clinical voice benchmarks. "Held-out" indicates whether a sealed test partition or honour-system test manifest is shipped; "LODO" indicates cross-corpus leave-one-dataset-out tasks.

Benchmark	#langs	#disease families	#conditions	#corpora	LODO	Harness
CogBench [Feng et al., 2025]	2	1 (cognitive)	1	2	no	partial
SLP-Helm [Patel et al., 2025]	2	1 (pediatric SLP)	1	2	no	yes
VoxClinBench (ours)	**4**	**2**	**12**	**5**	**yes (2 tasks)**	**yes**

Consortium corpora. The Bridge2AI-Voice consortium released v3.0 of its multi-condition English corpus on 2025-12-16 (PhysioNet DOI 10.13026/k81f-qr68) [Bensoussan et al., 2025, Goldberger et al., 2000] under PhysioNet's Credentialed Health Data License, shipping derived features (spectrograms, MFCC, EMA, PPG, prosody) rather than raw audio. Raw audio exists but is gated behind a controlled institutional application (DACO). The only published B2AI baseline, MARVEL [VoxClinBench Authors, 2026], predates the v3.0 psychiatric expansion, which is why MARVEL reports only eight physical conditions. VoxClinBench covers all twelve Tier 2 conditions and adds cross-lingual tasks so that the B2AI corpus can be audited against external cohorts without re-releasing restricted audio.

3 VoxClinBench Design

Disease-family decomposition is a first-principles design commitment, not a statistical convenience: physical and psychiatric conditions foreground different acoustic substrates (articulatory/phonatory kinematics vs prosodic/spectral affect), and clinical specialisation mirrors this distinction; scoring them jointly would conflate two different biomarker questions and two different measurement regimes. VoxClinBench therefore defines 18 tasks (16 within-corpus + 2 cross-lingual LODO) (task card in supp Table 3; results in Table 2) grouped into two sub-leaderboards: **Family A (physical)** and **Family B (psychiatric)**.

Within each family we run Holm–Bonferroni correction at α=0.05 independently, reflecting the disjoint measurement targets and preventing over-correction across families that ought not to compete.

Task inclusion rule. A task enters VoxClinBench iff (i) at least 30 positive subjects exist in the corpus, (ii) the label is clinician-confirmed or self-report validated against a standard instrument (e.g. PHQ-8 for depression), (iii) the split can be made subject-disjoint, and (iv) a public or credentialed license permits redistribution of the subject-ID manifest. We explicitly *exclude* B2AI's "prior psychiatric history" label because it is definitionally composite (overlaps depression, PTSD, and ADHD in the same cohort) and therefore not an independent biomarker task; including it would inflate leaderboard numbers without predictive content.

3.1 Corpora

Five corpora span four languages and two disease families. B2AI (English) contributes the largest cohort and the only psychiatric labels. NeuroVoz (Spanish) and SVD (German) provide external physical cohorts. E-DAIC (English, AVEC 2019 release; n=275, superset of the 189-subject DAIC-WOZ [Gratch et al., 2014] cohort + 86 new AVEC 2019 interviews) provides an external English depression cohort for within-corpus benchmarking. MODMA [Cai et al., 2022] (Mandarin, n=52) is retained as a *cross-lingual validation target only*: it is never used as a training source in v0.3 LODO rows, and its small sample size (n=52) keeps it outside the Holm family for ranking; zero-shot evaluation of English-trained probes on MODMA appears in Table 2 (A17) as a directional validation,

Table 2: VoxClinBench 18-task catalogue (16 within-corpus + 2 LODO) with two reference-model columns in parallel. Family A physical/motor/laryngeal/respiratory; Family B psychiatric. Shaded rows = **T2 diagnostic** ($n_{pos} < 50$; wide CIs, not decisive for ranking); unshaded rows = **T1 robust**. MARVEL column: 3-seed seasoned leak-free rerun, WavLM-Large + LogReg frozen [VoxClinBench Authors, 2026], source `checkpoints/marvel_v3/tier2/seasoned_seed_{1,2,42}/seasoned_result.json` (stage1 `test_unified_per_disease`); MARVEL T2 macro-AUROC is 0.744 ± 0.008 on this run. 8-mod column: 5-seed 8-modality CNN–Transformer (§4), source `artifacts/modal_sync/v0_2_upgrade/eval_b2ai_5seed_rerun/results_5seed.json` and patient-level means/CIs from `artifacts/eda/bootstrap_ci_summary.json`. "—" = not run. Columns are listed for parallel inspection; no "Exceeds"/"Ties"/"Defines" verdicts or Δ labels are assigned in this table. External-corpus rows are 2–3-seed descriptive reports (supp §2; per-seed predictions archived, re-CI-computable). AUPRC for low-prevalence tasks (A8/A10/A11) in supp.

#	ID	Corpus / Lang / Task	n_{pos}/n_{neg}	Metric	MARVEL	8-mod reference
Family A — Physical / Motor / Laryngeal / Respiratory						
A1	b2ai.parkinsons	B2AI / en / PD	73/689	AUROC+AUPRC	0.873±0.019	0.874±0.015
A2	b2ai.airway_stenosis	B2AI / en / iSGS	55/707	AUROC+AUPRC	0.909±0.014	0.916±0.006
A3	b2ai.vf_paralysis	B2AI / en / VFP	60/702	AUROC+AUPRC	0.844±0.033	0.959±0.016
A4	b2ai.laryngeal_dystonia	B2AI / en / SD-LD	50/712	AUROC+AUPRC	0.820±0.056	0.947±0.013
A5	b2ai.cognitive_impair†	B2AI / en / MCI	45/717	AUROC+AUPRC	0.927±0.025	0.913±0.013
A6	b2ai.benign_lesions	B2AI / en	58/704	AUROC+AUPRC	0.771±0.050	0.834±0.030
A7	b2ai.mtd	B2AI / en / MTD	52/710	AUROC+AUPRC	—	0.703±0.028
A8	b2ai.chronic_cough†	B2AI / en	41/721	AUROC+AUPRC	—	0.525±0.022
A13	neurovoz.parkinsons	NeuroVoz / es / PD	53/55	AUROC	—	0.961
A14	svd.vf_paralysis	SVD / de / VFP	213/300	AUROC	—	0.929
Family B — Psychiatric						
A9	b2ai.depression†	B2AI / en	44/718	AUROC+AUPRC	—	0.700±0.125
A10	b2ai.ptsd†	B2AI / en	38/724	AUROC+AUPRC	—	0.707±0.033
A11	b2ai.adhd†	B2AI / en	47/715	AUROC+AUPRC	—	0.619±0.060
A12	b2ai.psychiatric_history†	B2AI / en	—	AUROC	—	0.708±0.037
A16a	edaic.depression	E-DAIC / en / PHQ≥10	82/193	AUROC	—‡	0.600±0.075
A16b	edaic.phq8_regression	E-DAIC / en / PHQ-8	275 tot	Pearson/CCC	—	r=0.196; CCC 0.188
A17	modma.depression†§	MODMA / zh / MDD (zero-shot cross-lingual target)	23/29	AUROC	—	0.799±0.016
Cross-lingual LODO						
B1	lodo.pd.test_neurovoz	B2AI-en → NeuroVoz-es	—	AUROC	—	see A13
B6	lodo.vfp.test_svd	B2AI-VFP-en → SVD-de	—	AUROC	—	open‡

†T2 diagnostic. §A17 is a *cross-lingual validation target* (n=52 MODMA Chinese, zero-shot). Training pool: E-DAIC (n=275 EN subjects; DAIC-WOZ ⊂ E-DAIC, so the former v0.2 "DAIC+E-DAIC" pool is relabelled "E-DAIC" with no unique-subject loss). Value 0.799 ± 0.016 is 5-seed WavLM-Base L9 frozen + MLP head, source `artifacts/modal_sync/sequential_fewshot_20260410/`. MODMA is outside the Holm family and not ranked against Family B tasks; n=52 limits the directional claim to "English-trained probe transfers above chance to Chinese zero-shot on one cohort". ‡A16a MARVEL cell is withdrawn: all prior MARVEL E-DAIC values came from a LODO protocol that placed DAIC-WOZ ⊂ E-DAIC subjects in both train and test (189/189 leak); no clean within-E-DAIC MARVEL value is available at this release. The 8-mod column here is the within-dataset WavLM-Base L9 frozen + LogReg value (5 seeds, 80/20 subject-disjoint, source `artifacts/modal_sync/v0_3/eval_edaic_gl_vs_raw/results.json` raw arm). B6 open; cross-corpus gap attributable to language/cohort shift per §6(c) (GL ruled out).

not a scored ranking. DAIC-WOZ is not listed as a separate corpus because its participant IDs are a strict subset of E-DAIC (189/189 overlap, verified 2026-04-14); any within- or across-corpus evaluation that treats them as independent would be subject-leaked. Earlier releases of this benchmark (v0.2) did so and are retracted in supp §6. Full corpus provenance, license, and access-control protocol are in supp §3 and §4. We use the 2025-12-16 public release of B2AI v3.0 [Bensoussan et al., 2025, Goldberger et al., 2000].

3.2 Task catalogue

3.3 Protocol

Splits. Every task uses a subject-level stratified $80/20$ train/test split; within train, a further stratified $80/20$ train/val split. No subject crosses splits. Canonical JSON split manifests for all 18 tasks are

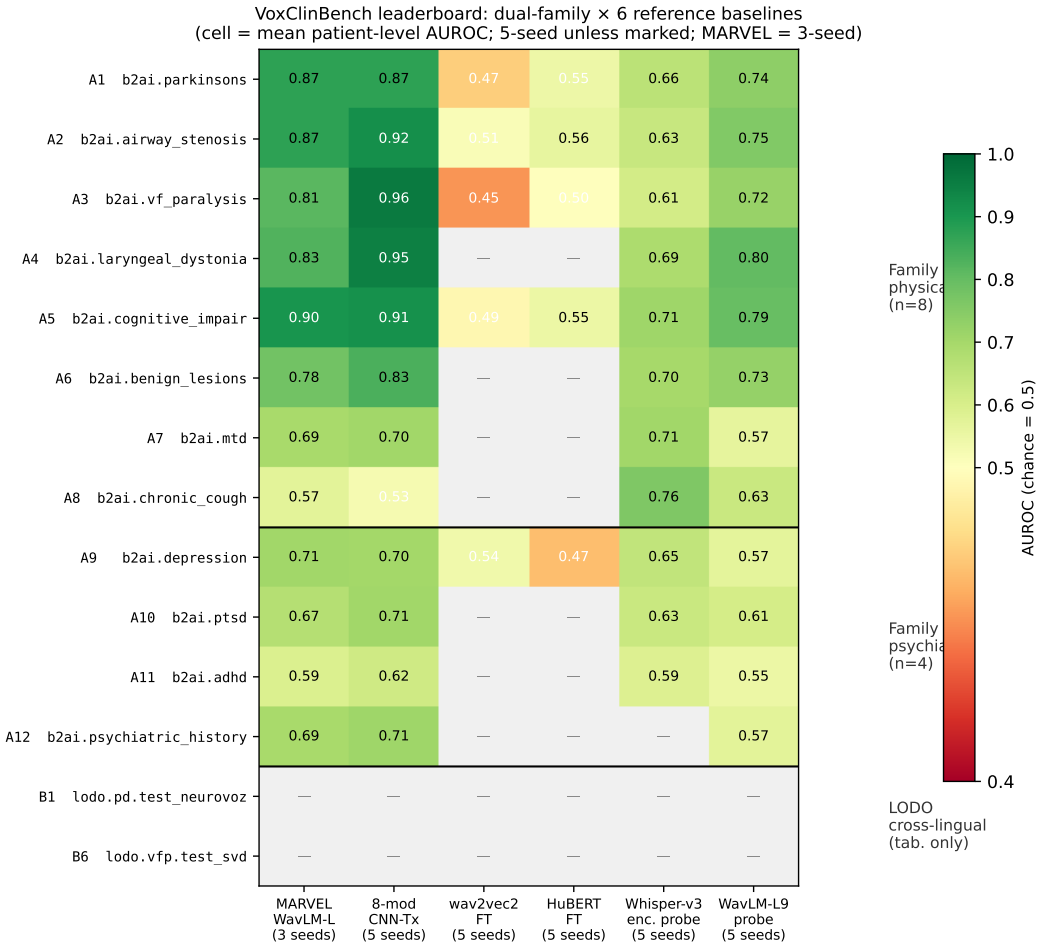

Figure 1: VoxClinBench dual-family leaderboard. Rows group Family A (8 physical voice disorders; macro_physical_8 $= 0.834$, SD 0.008) and Family B (4 psychiatric conditions; macro_psychiatric_4 $= 0.684$, SD 0.061) under the shared subject-disjoint protocol, with LODO rows B1/B6 reserved (tabulated only in Table 2). Columns are six reference baselines evaluated in parallel: MARVEL WavLM-Large probe (3 seeds), 8-mod CNN-Transformer (5 seeds), wav2vec2-base FT (5 seeds, Family A subset only), HuBERT-base FT (5 seeds, same subset), Whisper-large-v3 encoder probe (5 seeds, best-layer per disease), and WavLM-L9 frozen probe (5 seeds). Cells report mean patient-level AUROC; "—" marks baselines not run on that disease (e.g. wav2vec2/HuBERT FT were fielded on the 5-disease CLGP tail for the GL-small-N finding). Colour is a diverging scale centred at chance (0.5); 5-seed patient-level bootstrap CIs are in supp §2. The figure visualises the CLGP dual-family spine: no baseline dominates both families simultaneously, and the physical/psychiatric gap (≈ 0.15 AUROC on 8-mod) is what the benchmark is built to expose.

shipped in v0.2 under `voxbench/splits/` (`{task}.seed{s}.{train,val,test}.json`), generated deterministically by `scripts/eval_*.py` from a fixed corpus snapshot and cross-validated against the per-subject probability archives released alongside this paper. Access control is enforced at the *data-fetch* layer, not the submission layer: `voxbench fetch` verifies PhysioNet, USC/ICT, and SVD credentials before audio is downloaded (§7, Release). A hosted evaluation server accepting per-subject probability files is targeted for v1.0; v0.2 submissions self-report under the shipped manifests with *mandatory* code and per-seed prediction release so an external party can re-execute scoring end-to-end.

Seeds. Leaderboard rankings use the uniform 5-seed B2AI Tier 2 rerun $\{0, 1, 2, 3, 4\}$ covering all SSL FT/probe baselines (wav2vec2, HuBERT, Whisper-large-v3); external-corpus entries (A13/A14) and depression LODO use 3 seeds as descriptive reports outside the Holm family. Per-seed raw

predictions for all 18 tasks are archived so CIs can be recomputed under any matched-seed subset (two minor regime exceptions itemised in supp §2). The evaluator draws B=2000 subject-level percentile bootstrap resamples per test set for 95% CIs.

Task inclusion. The $n_{pos} \geq 30$ admission threshold is the minimum for a task to enter the leaderboard; our internal reporting focus remains on Tier 1+2 conditions (≥ 40 positive subjects) where single-cohort variance is tractable. Tasks with $30 \leq n_{pos} < 40$ ship as diagnostic entries whose CIs are honestly wide.

Metrics. Primary metric is subject-level AUROC for binary tasks and Pearson r + CCC for the single regression task (A16b, v0.2). AUPRC is reported as a secondary metric for tasks with positive prevalence $< 10\%$ (A8 chronic cough, A10 PTSD, A11 ADHD) in supp §2; Table 2 lists AUROC as the headline metric and tags AUPRC-reportable rows. Between-submission comparisons use the DeLong U-statistic test (`voxbench compare --test delong`) [DeLong et al., 1988, Sun and Xu, 2014], implemented in v0.2 per the Sun–Xu (2014) fast $O(N \log N)$ kernel and verified against 7 unit tests covering agreement with the paired subject-level bootstrap (B=2000) which remains available as `--test paired-bootstrap`. Either test is used only for within-cohort comparisons (both scores drawn from the same test subjects); cross-cohort deltas are reported as directional.

Correction. Within each family (A, B) we apply Holm–Bonferroni at α=0.05 against the MARVEL reference submission on the subset of tasks that admit a within-cohort MARVEL comparator. Family A (physical) has m=8 B2AI within-cohort tasks (A1–A8); Family B (psychiatric) has zero within-cohort MARVEL comparators at this release and is reported descriptively (no Holm family). External-cohort tasks (A13 NeuroVoz, A14 SVD) and LODO tasks (B1, B6) are reported descriptively; they do not enter the Holm family because no published same-protocol MARVEL number exists. Families are corrected independently to avoid over-correcting across disjoint measurement targets; joint correction across A and B is available as an optional `--family joint` flag in the evaluator.

Per-cell reporting. Table 2 lists reference models in parallel columns. No "beats" / "ties" / "Δ" label is assigned in the table. Within-family Holm-adjusted DeLong p-values against the MARVEL column (where a same-protocol MARVEL cell exists) are reported in supp §9. Both columns are reported for parallel inspection; the benchmark does not adjudicate which is preferred for clinical deployment.

Hyperparameters, SSL layers, tracks. Hyperparameters are fixed from preliminary experiments (no grid search on benchmark splits); target SSL layers (WavLM L9, XLSR-53 L18, HuBERT L9) are chosen on subject-disjoint validation splits within each training fold. Two tracks are scored separately: *frozen features* and *full fine-tune*; submissions declare the track.

3.4 Leaderboard governance & release

Splits are released as subject-ID manifests under CC BY 4.0; code is MIT. Raw audio is *not* redistributed; `voxbench fetch` verifies PhysioNet credentialing, USC/ICT EULAs, and SVD access forms before downloading from official sources. New corpora may be admitted by versioned PR; a new version bumps the minor number and publishes a new manifest set. Deprecated tasks (e.g. those whose upstream corpus license changes) are marked in the manifest with a grace period of one minor release before removal. Per-task submissions emit one JSON with fields {`task_id`, `model_name`, `seed`, `auroc`, `auprc`, `ci95`, `n_test`, `subject_probs`, `config`}; the evaluator recomputes AUROC, AUPRC, bootstrap CI, and Holm-corrected flags from `subject_probs` so submitters cannot game CIs by reporting only summary statistics. The per-submission protocol (artefact format, CLI, manifest schema, rejection criteria) is detailed in supp §12.1. A submission ships per-task, per-seed prediction CSVs (`subject_id,predicted_prob`) plus a `manifest.json`; the evaluator recomputes AUROC/AUPRC/CI/Holm flags. Full CLI surface, metric spec, tie-breaking, and rejection criteria are in supp §12.1.

4 Reference Baselines & Griffin–Lim Bridge

VoxClinBench ships a panel of reference models, reported in parallel, with no single model promoted as "ours". **MARVEL** [VoxClinBench Authors, 2026] extracts WavLM-Large mean-pooled embeddings and runs logistic regression on top, frozen (`class_weight="balanced"`); we re-run it on every VoxClinBench task under the shared protocol. The v0.2 leak-

free 3-seed rerun of MARVEL gives a B2AI Tier-2 macro-AUROC of 0.744 ± 0.008 (`artifacts/modal_sync/seasoned_leakfree_20260407/`).

We also include an **8-modality CNN–Transformer** fusion variant we constructed (full architecture and training details in supp §1).

Griffin–Lim bridge. B2AI's public v3.0 tier distributes magnitude spectrograms, not raw audio. We unlock SSL-native pipelines on this tier by Griffin–Lim inversion at 16 kHz [Griffin and Lim, 1984] (parameters in supp §1) and feeding the reconstructed waveform to a frozen WavLM. On external corpora where both raw and reconstructed audio are available, WavLM-L9 cosine similarity between GL-reconstructed and raw embeddings is 0.975 on E-DAIC (supp §4); a v0.3 paired GL-vs-raw study shows the WavLM-L9 probe AUROC matches or modestly exceeds raw on both a prosodic task (E-DAIC depression, $\Delta = +0.018$, N=275) and a phase-sensitive task (SVD voice pathology, $\Delta = +0.032$, N=513, p<0.001, 5 seeds each; `artifacts/modal_sync/v0_3/eval_{edaic,svd}_gl_vs_raw/results.json`), ruling out GL reconstruction as the bottleneck for the SSL pipeline; see §6(c) for the small-N reframing.

5 Results

Dual-family headline. Under the shared 5-seed subject-disjoint protocol, the 8-modality reference baseline on B2AI Tier-2 posts macro-physical-8 $= 0.834 \pm 0.008$ (95% CI $[0.791, 0.873]$) and macro-psychiatric-4 $= 0.684 \pm 0.061$ ($[0.553, 0.805]$); combined macro-12 $= 0.784 \pm 0.020$ is reported for continuity but is not the primary headline (see §6 for the family-separation rationale). Five physical diseases exceed 0.87 AUROC — VF paralysis (0.959 ± 0.016), laryngeal dystonia (0.947 ± 0.013), airway stenosis (0.916 ± 0.006), cognitive impairment (0.913 ± 0.013), Parkinson's (0.874 ± 0.015) — while the four psychiatric conditions sit at 0.619–0.708 with substantially larger across-seed variance reflecting small n_{pos} and self-report label noise. Figure 1 reports the same operating point alongside MARVEL, wav2vec2-FT, HuBERT-FT, Whisper-large-v3 encoder, and WavLM-only probe columns in parallel; no single reference model dominates across all 12 diseases. Per-disease 95% CIs, DeLong p-values, and per-seed predictions are in supp §2 and supp §9.

B2AI Tier 2. Neither reference dominates the 12-disease panel; two phonation tasks (MTD 0.703 ± 0.028, chronic cough 0.525 ± 0.022) remain below 0.75 AUROC across every shipped reference model, attributable to small-N not GL (§6(c)). The bundled reference panel (8-mod CNN–Transformer, wav2vec2/HuBERT FT, Whisper-large-v3 encoder, MARVEL WavLM-Large probe, WavLM-only L9 probe at 0.669 ± 0.016 macro-12) documents the currently-achievable operating point; future submissions may substitute stronger backbones under the same protocol. The full-FT collapse on reconstructed B2AI audio is treated as a benchmark-level finding in §6(d).

Within-dataset depression (E-DAIC, Family B): the clean WavLM-Base L9 frozen + LogReg probe value is 0.600 ± 0.075 (5 seeds, 80/20 subject-disjoint, PHQ-8$\geq$10; raw arm of `eval_edaic_gl_vs_raw/results.json`). No MARVEL comparator is reported on E-DAIC because all prior MARVEL E-DAIC values came from a LODO protocol that placed DAIC-WOZ $\subset$ E-DAIC subjects in both train and test (189/189 leak; see supp §6); DAIC-WOZ is retired as a separate corpus. The v0.2 MODMA value 0.799 ± 0.016 is retained as a zero-shot cross-lingual validation target (A17, n=52); the v0.2 MODMA-in-training LODO rows (B2, B4, B5) remain retracted.

Cross-lingual rows A13 (NeuroVoz PD, 0.961) and A14 (SVD VFP, 0.929) are reported in Table 2. Per-seed AUROC, AUPRC, and bootstrap CIs are in supp §2.

6 Benchmark-level Findings

A shared protocol makes four patterns legible that no single-corpus paper could see.

(a) Language-matched SSL shows higher AUROC than English-only SSL on non-English tasks. On NeuroVoz (es), HuBERT-L9 reaches 0.961 AUROC; XLSR-53-L18 reaches 0.955 ± 0.045; English-only WavLM-Base-L9 reaches 0.955 ± 0.041 (3-seed, matched-protocol; supp §2.2). On SVD (de), XLSR-53-L18 reaches 0.929 ± 0.021 vs. WavLM-Base-L9 0.885 ± 0.022. XLSR-53 matches WavLM-Base on NeuroVoz (≈ 0.955) and exceeds it by $+0.044$ on SVD; the advantage is

corpus-dependent, not uniform. Cohort-age heterogeneity (NeuroVoz median 68y, SVD median 23y) is an upstream recruitment confound that per-corpus AUROC should be read against (supp §8).

(b) Cross-lingual disease transfer is disease-dependent. Parkinson's and VF paralysis are both phonation disorders, yet the paired cross-lingual rows behave differently under matched zero-shot protocol. B2AI-EN PD $\rightarrow$ NeuroVoz-ES PD reaches 0.850 ± 0.006 AUROC with a full ensemble (vs MARVEL zero-shot 0.528); EMA (articulatory) transfers at 0.628 while MFCC (spectral) collapses to 0.406, consistent with PD motor symptoms manifesting in language-invariant articulatory features. VF paralysis transfers poorly between English B2AI and German SVD: a WavLM-L9 probe trained on either drops to ≈ 0.46 on the other (B2AI$\rightarrow$SVD 0.461; SVD$\rightarrow$B2AI 0.445; supp §7.3); joint B2AI+SVD training recovers SVD to 0.878 but degrades within-B2AI from 0.698 to 0.628. Depression follows the PD pattern: an E-DAIC-trained WavLM-L9 probe transfers zero-shot to Chinese MODMA at 0.799 ± 0.016 (A17, n=52), a single-cohort positive-transfer point. PTSD/ADHD/psychhistory have no external non-English cohort at this release and remain cross-lingual unknowns.

(c) Griffin–Lim reconstruction is not necessarily a bottleneck on the corpora tested. A recurring concern on spectrogram-only releases is that GL phase-loss imposes a downstream-AUROC ceiling, plausibly explaining the 0.49–0.58 plateau on B2AI benign lesions/MTD/chronic cough. We test this by training a WavLM-Base L9 frozen probe on identical subjects, audio range, and seeds, swapping only raw-vs-GL audio.

Table 3: Paired GL-vs-raw causal study, WavLM-Base L9 probe, 5 seeds, subject-disjoint 80/20. GL matches or modestly exceeds raw in both prosodic and phase-sensitive domains. Sources: `artifacts/modal_sync/v0_3/eval_edaic_gl_vs_raw/results.json` and `eval_svd_gl_vs_raw/results.json`.

Dataset	Task	N	raw	GL	Δ (GL$-$raw)	paired p
E-DAIC	depression binary	275	0.600 ± 0.075	0.618 ± 0.071	$+0.018$	0.08
SVD	voice pathology	513	0.831 ± 0.026	$\mathbf{0.863 \pm 0.010}$	$\mathbf{+0.032}$	$\mathbf{< 0.001}$

On the two corpora where paired raw audio is available (SVD and E-DAIC), GL reconstruction does not degrade WavLM-L9 probe AUROC. This reduces, but does not rule out, concerns for other B2AI conditions where no raw audio is available for paired comparison; the B2AI plateau on benign lesions/MTD/chronic cough is therefore plausibly attributable to small-N ($n_{\text{pos}} \in [41, 58]$, $\approx$ 5–12 positives per 126-subject test fold) rather than reconstruction. The jitter/shimmer/AVQI collapse and HiFi-GAN-spike non-improvement (supp §7.2, §2.4) are consistent with this picture: hand-crafted phase-perturbation features are inappropriate inputs for the GL pipeline, but SSL probes on SVD/E-DAIC are not degraded.

(d) Full end-to-end fine-tuning collapses on small-N reconstructed audio. Under the naïve reading of the SSL literature, full fine-tuning should dominate frozen probes; the benchmark shows the opposite. wav2vec2-base and HuBERT-base end-to-end FT on GL-reconstructed B2AI audio ($n_{\text{pos}} \in [38, 73]$, LP-FT 3+3 epochs, 5 seeds) produces monotonically decreasing loss but held-out AUROC collapses to 0.45–0.56 across five Tier-2 conditions, while the frozen 8-modality stack on identical subjects reaches 0.79 macro [Baevski et al., 2020]. Without a matched frozen-probe comparator, an isolated FT study reporting 0.45–0.56 would read as "this task is hard" rather than as catastrophic forgetting under small-N — the kind of disagreement shared protocols are designed to surface.

7 Limitations & Release

Limitations. Wide CIs on weak B2AI diseases ($n_{pos} \in [41, 58]$); psychiatric training corpora are English-only (B2AI, E-DAIC), with MODMA (n=52 Chinese) supplying a small-sample cross-lingual validation target only; GL-bridge gap bounded $\Delta \in [+0.018, +0.032]$ (§6(c)); LODO substitutes for cross-cohort Δs against FEMH/DVBIC/ADReSSo [Koudounas et al., 2024, Marmar et al., 2019, Pan et al., 2021]; demographics preliminary (supp §8). **Release.** Anonymous mirrors under slug `voice-bench-submission/voxclinbench` ship harness, preprocessing, model code,

66 per-seed prediction CSVs, Croissant, datasheet (MIT / CC BY 4.0; raw audio not redistributed; licenses and maintenance in supp §14).

8 Ethics & Broader Impact

Intended use. VoxClinBench is a *research-grade* benchmark for comparing acoustic representations on clinical voice screening tasks; it is not a diagnostic device, is not clinically validated, and must not substitute for a licensed clinician. Per-task AUROC values are population-level ranking statistics, not individual risk scores.

Positive societal impact. Voice is cheap and non-invasive, making acoustic screening an attractive *pre-referral* aid in settings with limited specialist access. Consolidating five corpora across four languages (en, es, de, zh) behind one harness makes it possible to measure whether English methods generalise to Spanish PD (NeuroVoz), German voice pathology (SVD), and Chinese depression (MODMA) populations typically excluded from English-only benchmarks.

Risks of harm. (i) *Small-N variance*: B2AI tail diseases have $n_{\text{pos}} \in [41, 58]$ and MODMA has only $N{=}52$ (23 MDD / 29 HC); point estimates are directional. (ii) *Demographic imbalance*: race/ethnicity is recorded only for B2AI (76% White, 8.5% Black, 5.8% Asian) and unavailable for the four non-US cohorts; age distributions are cohort-skewed (NeuroVoz median 68 y, SVD median 23 y); sex coverage is $\geq 99\%$ for B2AI/NeuroVoz/E-DAIC/MODMA but 48% missing for SVD; MODMA is 69% male (supp §8). Any "language-matched SSL" effect is therefore confounded with age and sex. (iii) *Unrepresented languages*: four languages are a narrow slice and do not license claims about tonal languages other than Mandarin, non-Indo-European families, or low-resource variants. (iv) *Voice as biometric*: raw speech is re-identifiable; we redistribute only the harness, preprocessing code, per-seed prediction CSVs, and Croissant/datasheet metadata; raw audio remains with each source provider under its DUA.

Consent, provenance, licensing. Each corpus is redistributed strictly under its upstream licence and access tier (supp Table 5). We do not re-release raw recordings or any artefact that re-identifies a participant. Labels derive from clinical diagnosis for four corpora and from a PHQ-8 cutoff ≥ 10 for E-DAIC; psychiatric labels are *screening* labels, not clinical diagnoses, and must not be surfaced back to a participant as such. DAIC-WOZ was retired after discovering a 189/189 subject overlap with E-DAIC to prevent train/test leakage; the retraction is logged in the supplementary datasheet (§14).

Fairness, misuse, and maintenance. B2AI is US-cohort-dominant; NeuroVoz skews elderly Spanish PD; SVD skews young German HC; MODMA is a single Chinese site with $n{=}52$. A model that wins on this benchmark has *not* been shown to work across the demographic strata required for deployment. We discourage: (a) citing per-task AUROC as deployment-readiness, (b) inferring individual diagnoses from aggregate performance, and (c) applying pretrained checkpoints outside the benchmark's coverage without site-specific recalibration and clinician oversight. VoxClinBench is versioned (v0.2); any discovered label error, subject overlap, or leakage triggers a versioned retraction in the datasheet with prior versions pinned for reproducibility.

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